LINEAR ALGEBRA ON ONE PAGE

The LCM method as Gaussian elimination

Matching coefficients with the LCM is one row operation on the augmented matrix, repeated down the columns.

1 / THE RULE AND THE PICTURE

Write the system as a matrix, and each LCM step becomes one row operation

$$\left(\begin{array}{cc|c} 3 & 2 & 16 \\ 2 & 5 & 19 \end{array}\right) \xrightarrow{R_2 \leftarrow 3R_2 - 2R_1} \left(\begin{array}{cc|c} 3 & 2 & 16 \\ 0 & 11 & 25 \end{array}\right) \xrightarrow{\text{back-substitute}} \left(\begin{array}{cc|c} 1 & 0 & \frac{42}{11} \\ 0 & 1 & \frac{25}{11} \end{array}\right)$$

$3x + 2y = 16,\ 2x + 5y = 19$ $\qquad$ $11y = 25$ $\qquad$ $x = \frac{42}{11},\ y = \frac{25}{11}$

The coefficients sit left of the bar and the constants right of it:

$$R_2 \leftarrow \frac{L}{d}R_2 - \frac{L}{a}R_1$$

with $L = \mathrm{LCM}(a, d)$ the first column of row 2 becomes 0.

2 / THE THREE ALLOWED MOVES / NONE OF THEM CHANGES THE SOLUTION

Scale a row, add a multiple of one row to another, or swap two rows

row operation	what it does to the lines	in the LCM method
$R_i \leftarrow kR_i$	rewrites a line, as $6x + 4y = 32$ is $3x + 2y = 16$	multiply by L/a or L/d
$R_i \leftarrow R_i + kR_j$	replaces a line by another through the crossing point	subtract the matched rows
$R_i \leftrightarrow R_j$	relabels the lines	put the easier row on top

Every move keeps the common point of the two lines, which is why the final matrix has the same solution as the first. The two multipliers 3 and 2 in the picture are $6/2$ and $6/3$ with $L = \mathrm{LCM}(3, 2) = 6$, and the surviving pivot $11 = ae - bd$ is the determinant.

3 / ECHELON FORM AND BACK-SUBSTITUTION / FINISHING THE JOB

Forward elimination makes a staircase, and back-substitution climbs it

From $11y = 25$ we get $y = \frac{25}{11}$, and the first row $3x + 2y = 16$ gives $3x = 16 - \frac{50}{11} = \frac{126}{11}$, so $x = \frac{42}{11}$.
Continuing to divide the rows by their pivots and clear above them (Gauss–Jordan) leaves the identity on the left, with the solution in the right-hand column.

$$x = \tfrac{42}{11} \approx 3.82, \qquad y = \tfrac{25}{11} \approx 2.27$$

When the coefficients share a factor the LCM is smaller than the product: for $4x + 3y = 10$ and $6x + 5y = 16$ take $L = \mathrm{LCM}(4, 6) = 12$, so $R_2 \leftarrow 2R_2 - 3R_1$ turns $(6, 5 \mid 16)$ into $(0, 1 \mid 2)$, giving $y = 2$ and then $x = 1$, with multipliers 2 and 3 where the product 24 would have needed 4 and 6.

4 / WHEN THE STAIRCASE BREAKS / A ZERO ROW

A row of zeros means infinitely many solutions, and $0 = 1$ means none

system	operation	result	meaning
$6x + 4y = 20,\ 9x + 6y = 30$	$L = 18$: $R_2 \leftarrow 2R_2 - 3R_1$	$\left(\begin{array}{cc\|c} 6 & 4 & 20 \\ 0 & 0 & 0 \end{array}\right)$	the same line
$2x + y = 3,\ 4x + 2y = 7$	$R_2 \leftarrow R_2 - 2R_1$	$\left(\begin{array}{cc\|c} 2 & 1 & 3 \\ 0 & 0 & 1 \end{array}\right)$	parallel, $0 = 1$

5 / THREE UNKNOWNS / THE SAME MOVE, APPLIED TWICE

Clear the first column, then the second, and read z first

$x + y + z = 6$, $2x - y + z = 3$, $x + 2y - z = 2$. The moves $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - R_1$ give rows $(0, -3, -1 \mid -9)$ and $(0, 1, -2 \mid -4)$, and then $R_3 \leftarrow 3R_3 + R_2$ gives $(0, 0, -7 \mid -21)$, so $z = 3$; back-substitution gives $-3y - 3 = -9$, so $y = 2$, and then $x = 1$. Each equation is a plane, and the solution $(1, 2, 3)$ is the point where three planes meet.

ASIDE / WHY LCM

The LCM is the smallest multiplier that makes the coefficients equal, so the numbers in the matrix stay as small as possible.

ASIDE / CHECK IN PYTHON

`np.linalg.solve([[3,2],[2,5]],[16,19])` gives $3.818,\ 2.273$.

Convention. The augmented matrix lists coefficients then constants; $L = \mathrm{LCM}(a, d)$ is taken on the column being cleared.

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